

**An example
of
Projecting global sea-level
using
FACTS**

By Praveen Kumar

Intro to Installing FACTS

• What you will run today is a containerized (Docker), summer-school-specific build of FACTS and is **NOT the general distribution**.

- Runs on standard laptop, workstations and Desktops running macOS, Windows, and Linux.
- HPC setup is very different.
- Watch RAM and disk space at all times running the container
more samples or locations means much larger files.

Big experiments will crash a laptop. Start small and scale up.

Use the recipe to build the kitchen
(the container)
then cook (run FACTS) the
same way for everyone.

Today's Prerequisites:

Docker installed and running, ~30 GB free disk
space.

You can also install on a **flashdrive**.

**Git brings the
recipe**

source code + Dockerfile +
helper scripts



pkjr002

Before you Install:: READ

Software

- Docker Desktop (running)
- git + SSH key for GitHub
- Terminal (bash / zsh / WSL , IDE)

Hardware

- ~30 GB free disk
- ≥ 8 GB RAM (16 GB recommended)
- macOS / Windows / Linux
- Min 2 processors

Reference

- FACTS Quick Start
- Git code
- This slide deck

⚠ The HPC setup is structurally different. Don't try to lift this Docker workflow onto a cluster on your own — talk to the dev team.

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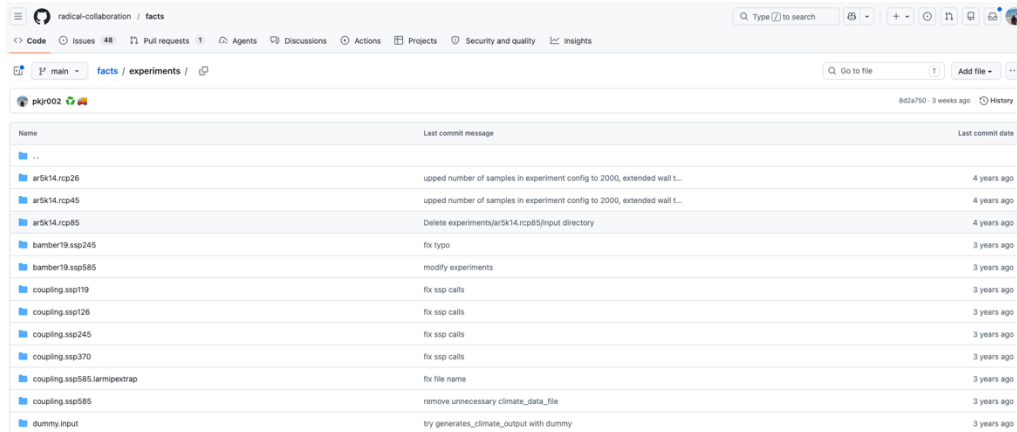
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Install FACTS

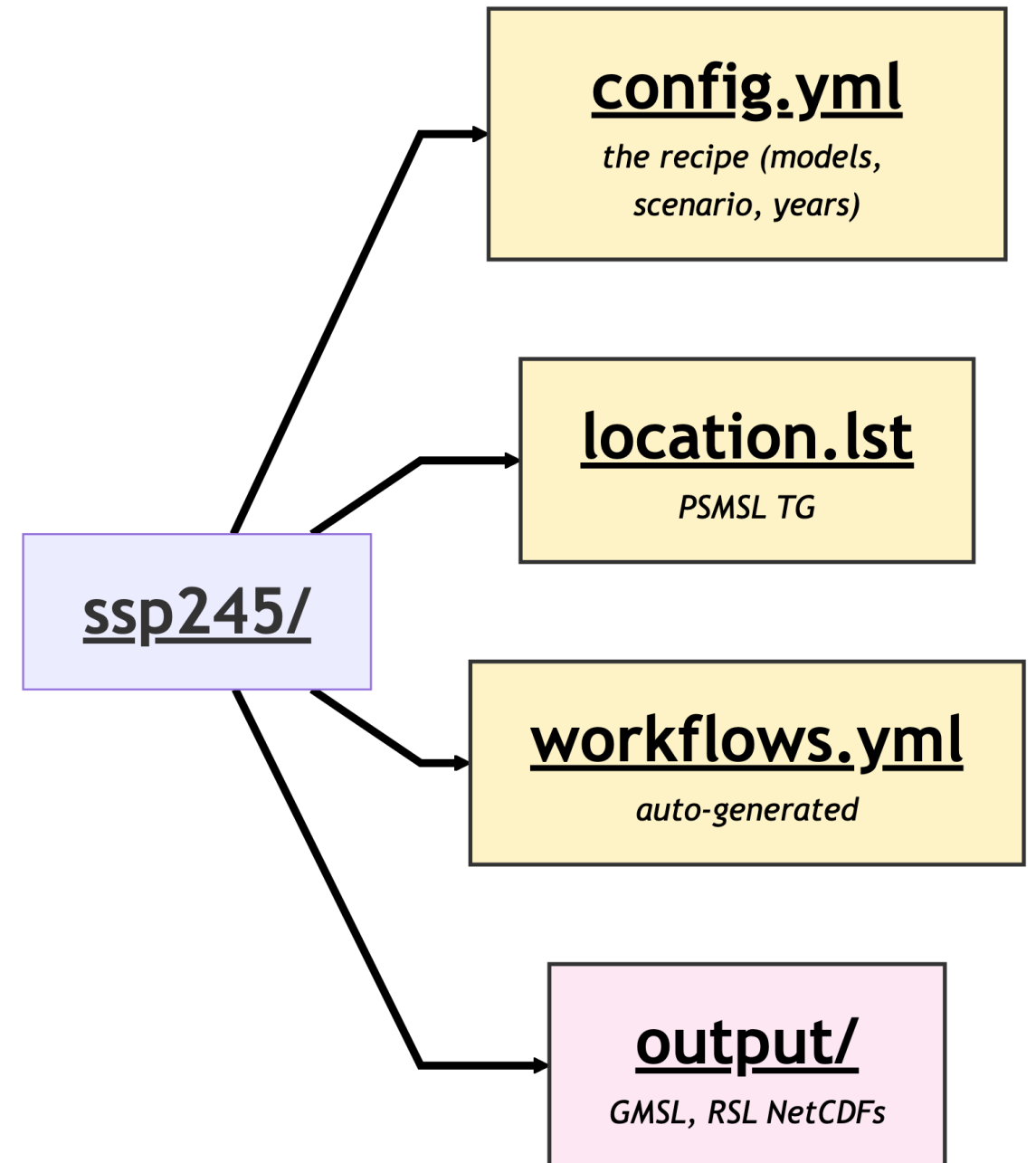
Use the guide Provided to

FACTS experiment

- A FACTS experiment (SSP245) is just a directory.
- Everything that defines the projection lives inside it.
- Run it by pointing runFACTS.py at the directory.



Name	Last commit message	Last commit date
..		
ar5k14.rcp26	upped number of samples in experiment config to 2000, extended wall t...	4 years ago
ar5k14.rcp45	upped number of samples in experiment config to 2000, extended wall t...	4 years ago
ar5k14.rcp85	Delete experiments/ar5k14.rcp85/input directory	4 years ago
bamber19.ssp245	fix typo	3 years ago
bamber19.ssp585	modify experiments	3 years ago
coupling.ssp119	fix ssp calls	3 years ago
coupling.ssp126	fix ssp calls	3 years ago
coupling.ssp245	fix ssp calls	3 years ago
coupling.ssp370	fix ssp calls	3 years ago
coupling.ssp585.larmipextrap	fix file name	3 years ago
coupling.ssp585	remove unnecessary climate_data_file	3 years ago
dummy.input	try generates_climate_output with dummy	3 years ago



FACTS Run

```
$ bash  
submit_docker_ssisls_experiment.sh  
ssp245
```

- I have configured FACTS to run with a submit script
- This is a very refined version of running the model.

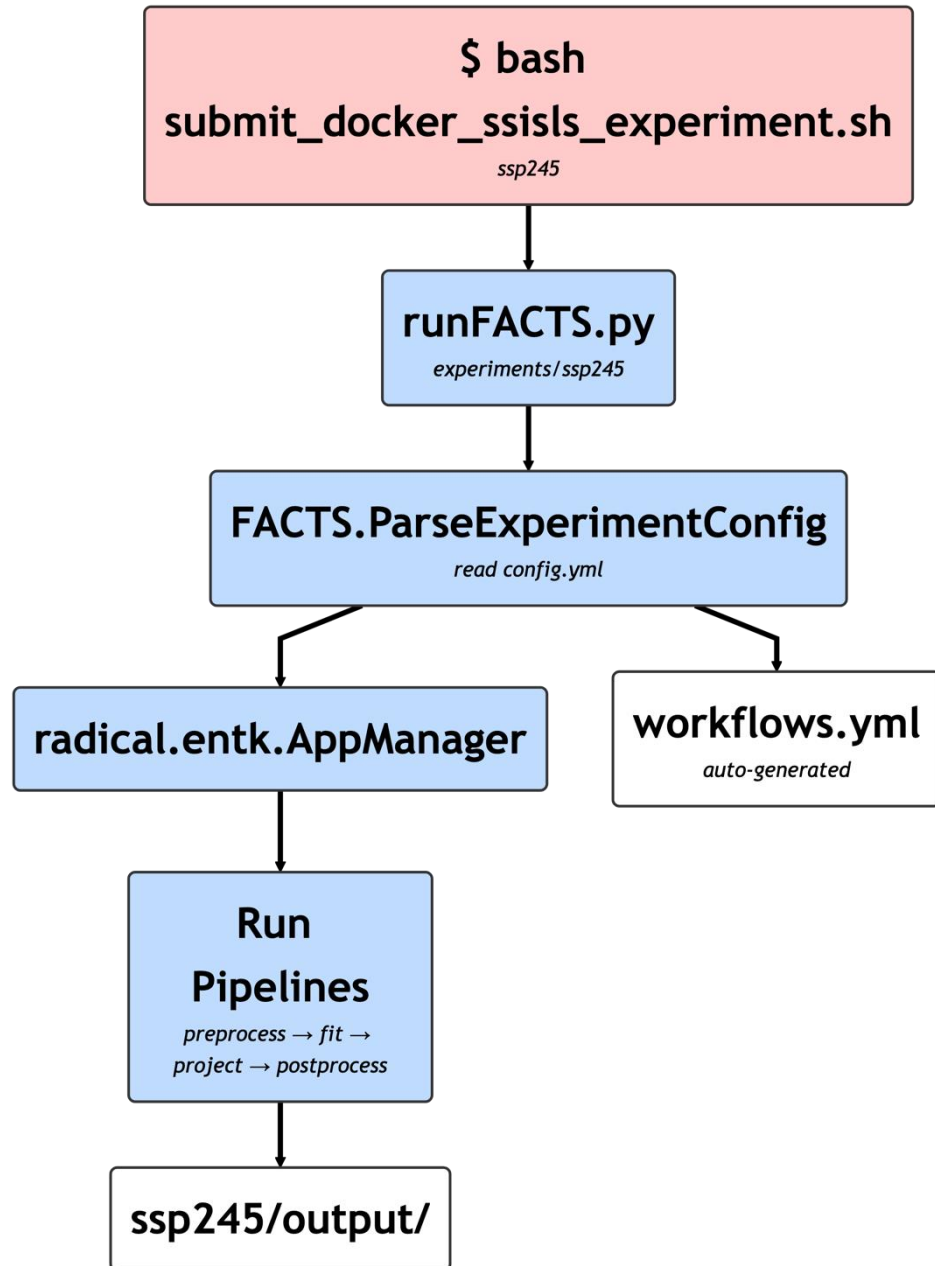
RADICAL Thinking

Research in Advanced Distributed Cyber- infrastructure and Applications Laboratory. Working at the intersection of computing and science to advance cyberinfrastructure tools.



[https://github.com/
radical-cybertools](https://github.com/radical-cybertools)





FACTS Run

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FACTS Experiment

Steps

Climate

One module is run during the climate step.

Within each **module** in an **component of sealevel**, stages are executed sequentially.

Stages
preprocess
fit
project
postprocess

Sea-level

module/pipeline
...

module/pipeline
...

module/pipeline
...

module/pipeline
...

module/pipeline
...

...

Multiple module representing different sea-level components can be run during sea-level step.

Workflow: Combination of modules run in sea-level step to be summed into a total projection of sea-level change.

Wf1:



Wf2:



Wf3:

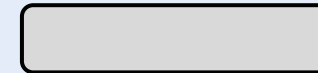
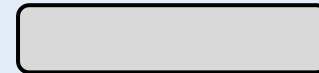
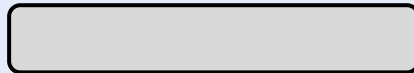


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Totaling



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config.yml

config.yml

```
global-options:
  nsamps: 2000
  scenario: ssp585
  pyear_start: 2020
  pyear_end: 2150
  pyear_step: 10
  baseyear: 2005

climate_step:          # produces GSAT / OHC for the SLR modules
  temperature: { module_set: fair, module: temperature,
                 generates_climate_output: true }

sealevel_step:          # the projection modules (the big one)
  GrIS1f: { module_set: FittedISMIP, module: GrIS,
            include_in_workflow: [wf1f, wf2f, wf3f] }
  deconto21: { module_set: deconto21, module: AIS,
               include_in_workflow: [wf3e, wf3f] }
  larmip: { module_set: larmip, module: AIS,
            include_in_workflow: [wf2e, wf2f] }
  ocean: { module_set: tlm, module: sterodynamics, include_in_workflow: [wf1f, wf1e, wf2e, ...] }
  ...

totaling_step: { facts/total · loop over workflows + scales }
esl_step:      { extremesealevel/pointsoverthreshold · loop over workflows }
```

workflows.yml — auto-generated

You can think of each workflow as a total sea-level from **summing over a specific combination of modules**

```
workflows.yml
(auto-generated)
wf1:
  global:
    - global/coupling.ssp585.emuAIS.emulandice.AIS_globalsl.nc
    - global/coupling.ssp585.emuGrIS.emulandice.GrIS_globalsl.nc
    - global/coupling.ssp585.emuglaciers.emulandice.glaciers_globalsl.nc
    - global/coupling.ssp585.ocean.tlm.sterodynamics_globalsl.nc
    - global/coupling.ssp585.lws.ssp.landwaterstorage_globalsl.nc
  local: []
  options:
    pyear_end: 2100

wf1f:
  global:
    - global/coupling.ssp585.GrIS1f.FittedISMIP.GrIS_GIS_globalsl.nc
    - global/coupling.ssp585.ar5glaciers.ipccar5.glaciers_globalsl.nc
  ...
```

What lands in output/ (NetCDFs)

python

```
# Canonical NetCDF schema for every output:
sea_level_change : float32 (samples=2000, years=14, locations=1)  units = mm
lat              : float64 (locations,)
lon              : float64 (locations,)
years            : int64   (years,)          # 2020, 2030, ..., 2150
locations        : int64   (locations,)      # PSMSL ids from location.lst

# Plotting at New York:
import xarray as xr
ds = xr.open_dataset('output/coupling.ssp585.total.workflow.wf3f.local.nc')
samples = ds['sea_level_change'].isel(locations=0)
q50, q05, q95 = samples.quantile([0.5, 0.05, 0.95], dim='samples')
```

... welcome to FACTS

The End